ICG - P2P

ASSIGNMENT - 3

Report

Date

Page

* **Q.1** Why does the notebook download imagery live from a server instead of shipping a satellite image file with the notebook? Give one practical reason:

→ The notebook downloads imagery live from the server because satellite images are very large & different users may require different locations & zoom levels. Downloading only the required area reduces storage usage and keeps the notebook lightweight & flexible.

* **Q.2** If you increased the zoom from 18 to 19, what happens to (a) the number of tiles downloaded and (b.) how clearly the model can see small rooftops?

→ If the zoom level increases from 18 to 19, the number of tiles downloaded increases because each tile covers a smaller area. The model can see small rooftops more ~~so~~ clearly because the image resolution becomes higher and rooftop details become sharper.

* **Q.3** We filter footprints with confidence >= 0.65. What would happen to our building count if we lowered this to 0.4? What if we raised it to 0.9? Which errors does each choice risk (missing real buildings vs. counting fake ones)?

Date
Page

→ If the confidence threshold is lowered to 0.4, the building count will increase because more detections are accepted. However, this may increase false positives & count fake buildings

→ If the confidence threshold is raised to 0.9, the building count will decrease because only very confident detections are accepted. This reduces false positives but may miss many real buildings.

→ Lower confidence risks counting fake buildings, while higher confidence risks missing real buildings.

*Q.5 Fine-tuning continues training from the sandbox weights instead of starting from scratch. Why is that faster & needs less data than training a brand-new model?

→ Fine-tuning is faster because the model already learned basic building features during previous training. Instead of learning everything again, it only adapts to Indian rooftop patterns. This requires less data & less computation compared to training a complitely new model from scratch.

Date
Page

* Q.6 Even after fine-tuning, best_Indian.pt only scores around 0.24 mAP on Indian data - far below the 0.7 - 0.8 it got in the sandbox. But the detection images look reasonable. How can the number be low while the picture looks good?

→ mAP is a strict mathematical metric that checks how closely the predicted boxes match the labelled ground truth boxes. Indian rooftops are dense, irregular, and sometimes poorly labelled, so even visually good detections may not perfectly match the labels. Therefore, the images can look reasonable while the numerical mAP score remains low.

* Q.7 We use the count of buildings per cluster as a "density score." For our next goal, what does a building count miss?

→ Building count alone does not represent actual population because all buildings are counted equally. A small one-storey house and a large multi-floor apartment tower are both counted as one building even though they can contain very different numbers of people. Therefore, building count cannot accurately estimate population density.

Date
Page

* **Q.8** You spent Session 1 training a model and Session 2 fine-tuning it — then we used Open Buildings for the real answer anyway. Was model-building a waste of time? Argue your view in 4-5 sentences.

→ Model-building was not a waste of time because it helped us understand how real-world machine-learning works. We learned about domain gap, fine-tuning, confidence thresholds, and geospatial processing. However, Open Buildings performed much better because it was trained on a massive dataset with large-scale resources. In real-world data science, using an existing high-quality dataset is often more practical than building a new model from ~~scratch~~ scratch. The assignment showed both the learning value of buildings ~~mo~~ models & the practical value of ~~ex~~ using existing tools.

CODE PART QUESTIONS :

Q.1 Why did the sandbox model fail on Lucknow even though it scored 0.7 - 0.8 mAP in Session 1?

→ The Sandbox model failed on Lucknow because it was trained mainly on US & European buildings, which are larger & more regular. Indian rooftops are smaller, denser, and more irregular, so the model could not generalize properly. This difference in data distribution is called the domain gap.

Q.2 Roughly how many buildings did your model detect in your patch? Where did it miss the most?

→ My model detected approx. 6455 buildings in the selected patch. The model missed the most buildings in dense urban & slum-like regions where rooftops were tightly packed, overlapping, and ~~irr~~ irregularly shaped.

Q.3 Open Buildings almost certainly found far more buildings than your model. Why? And given that - when would a real data scientist choose to use an existing dataset instead of running their own model?

→ Open Buildings found far more buildings because it was trained on extremely large datasets and optimized for large-scale detection across many regions. It handled dense Indian urban areas much better than the custom YOLO model. A real data scientist would choose an existing dataset when it is already highly accurate, reliable, and saves time and computational resources compared to building & training a new model from scratch.